

Liver Transplantation



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outline

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Definition:-

It is a partial or removal of the diseased liver and its replacement with a healthy liver in the same anatomic location. It is used to treat life threatening, end stage liver disease for which no other form of treatment is available.



Indications for liver transplantation:-

Liver transplantation is considered when the liver no longer functions adequately (liver failure). Some common causes of liver failure include:

- ❖ Sudden (acute) failure from infection or complications from certain medications
- ❖ Long-term failure as a result of the following conditions:-
 - Auto immune hepatitis.
 - Chronic hepatitis Hepatitis B and Hepatitis C (<5%)

Indications for liver transplantation:-

- Primary biliary cirrhosis. A rare condition where the immune system inappropriately attacks and destroys the bile ducts.
- Sclerosing cholangitis. Scarring and narrowing (Harding and thickening) of the bile ducts inside and outside of the liver. This results in backup of bile in the liver and liver failure.
- Biliary atresia. Malformation of the bile ducts. Common reason for transplant in children.
- Alcoholism which cause laennec's cirrhosis.

Indications for liver transplantation:-

- Wilson's disease. A rare inherited disease with abnormal deposition of copper throughout the body due to genetic mutation result in impaired hepatic excretion of copper including the liver, causing it to fail due to hepatitis and cirrhosis.
- Hemochromatosis (iron overload disease). A common inherited disease where the body is overwhelmed with iron due to the alteration of absorption of iron from intestine.
- Amyloidosis. The liver has abnormal deposits of an abnormal protein called amyloid that disrupts normal liver function
- Liver cancer.
- Thrombosis which in hepatic vein and blocks hepatic venous out flow.

Contraindications for liver transplantation:

- Sepsis.
- Malignancies outside the liver.
- Advanced hepatobiliary malignancy.
- Active substance abuse.
- Irreversible brain damage.
- Inability to comply with post-transplant immunosuppression regimen.
- AIDS.
- Active cardio pulmonary disease.

Determination the candidates for liver transplantation:

This interdisciplinary healthcare team may include the following professionals:

- Liver specialist (hepatologist).
- Transplant surgeons.
- Transplant coordinator, usually a registered nurse who specializes in the care of liver-transplant patients. This person will be primary contact with the transplant team
- Social worker to discuss your support network of family and friends, employment history, and financial needs.
- Psychiatrist to help you deal with issues, such as anxiety and depression, which may accompany the liver transplantation.
- Anesthesiologist to discuss potential anesthesia risks.
- Chemical dependency specialist to aid those with history of alcohol or drug abuse.
- Financial counselor to act as a liaison between a patient and his or her insurance companies.

Investigation before liver transplantation:

- Computed tomography, which uses X-rays and a computer to generate pictures of the liver, showing its size and shape.
- Doppler ultrasound to determine if the blood vessels to and from liver are open.
- Echocardiogram to help evaluate heart.
- Pulmonary function studies to determine lungs' ability to exchange oxygen and carbon dioxide.
- Blood tests to determine blood type, clotting ability, and biochemical status of blood and to gauge liver function. AIDS testing and hepatitis screening are also included.

How does the waiting list work?

- If the person become an active liver-transplant candidate, his name will be placed on a waiting list.
- Patients are listed according to blood type, body size, and medical condition (how ill they are).
- Each patient is given a priority score based on three simple blood tests (creatinine, bilirubin, and INR).
- The score is known as the MELD (model of end stage liver disease) score in adults and PELD (pediatric end stage liver disease) in children.
Patients with the highest

Medical care of the patient A waiting a liver transplant:

(A) Treatment of ascites

- Restriction of dietary sodium
- Diuretic therapy (spironolactone and furasemide).
- Paracentesis may be performed to remove ascitic fluid from the abdomen.
- A peritoneal venous shunt (laven shunt) may be surgically inserted which drains the abdominal cavity into a large central vein. This is not often used due to the significant risk of infection, DIC, or occlusion of the one way valve used.

Medical care of the patient A waiting a liver transplant:

Treatment of variceal bleeding:

- Fluid resuscitation
- Correction of coagulopathies
- Sclerotherapy to control hemorrhage
- Esophageal – variceal ligation (banding)
- Sengstaken – Blakemore tube:- compression device which works by balloon tamponade.

Treatment of variceal bleeding:

- Vasopressin or octreotide therapy
- TIPS.
- Distal splenorenal shunt (warren shunt): The distal splenic vein is anastomosed with the left renal vein. Portal venous circulation is still preserved but decompression of the gastric and esophageal varices occurs resulting in prevention of re bleeding.
- Partial mesocaval shunt: A small (< 15 mm) Dacron graft is placed between the superior mesenteric vein and inferior vena cava. This results in partial diversion of portal blood flow away from the liver

❖ **The donor**, who may be a blood relative, spouse, or friend, will have extensive medical and psychological evaluations to ensure the lowest possible risk. Blood type and body size are critical factors in determining who is an appropriate donor.

❖ **Recipients** for the living donor transplantation must be active on the transplant waiting list. Their health must also be stable enough to undergo transplantation with excellent chances of success.

What happens when they find a match?

When an organ has been identified a transplant coordinator will contact by telephone or by pager. Make sure that patient do not eat or drink anything once he have been called to the hospital. The transplant coordinator will notify of any additional instructions. When he arrive at the hospital, additional blood tests, an electrocardiogram, and a chest X-ray will generally be taken before the operation. Also may meet with the anesthesiologist and a surgical resident.

What happens during the transplant operation?

Liver transplants usually take from six hours to 12 hours. During the operation, surgeons will remove the liver and will replace it with the donor liver. Because a transplant operation is a major procedure, surgeons will need to place several tubes in body. These tubes are necessary to help body carry out certain functions during the operation and for a few days afterward:

During the operation, a tube will be placed through the mouth (endotracheal) to help breathe during the operation and for the first day or two following the operation. The tube is attached to a ventilator that will expand lungs mechanically.

What happens during the transplant operation?

Only certain transplant patients receive a T-tube, which remains in place for five months.

The tube causes no discomfort and does not interfere with daily activities.

- The bile is flushed from gall bladder to prevent damage to the walls of the biliary tract.
- Liver is perfused with a preservative and cooled.
- Immediate before translation liver is flushed with cold lactated ringer's solution to remove potassium and air bubbles.
- Anastomoses of the blood vessels and bile duct between the donor liver and the recipient's liver.

What happens during the transplant operation?

- Biliary reconstruction is performed with an end to end anastomoses of the donor and recipient common bile ducts.
- Liver Transplantation is a long surgical procedure, because the patient with liver failure has portal hypertension and many venous collateral vessels that must be ligated.
- Blood loss may be extensive during surgery.
- If shunt is performed:- should reverse to permit adequate portal venous blood supply to the new liver.

Indicators of post–Transplant liver functions

Neurological status:-

If the patient was encephalopathic pre-operative, should observe mental clearing as ammonia levels and cerebral edema decrease.

PT / PTI:-

The patient may have been coagulopathic pre-operative but these lab values should trend towards normal with out intervention if the liver is producing clotting factors.

Liver function tests:-

Albumin, AST, ALT, GGT, LDH, Alkaline phosphatase, total bilirubin, and NH₃ all begin trending toward normal if the grafted liver is functioning well.

Indicators of post-Transplant liver functions

Bile production:-

The newly grafted liver should begin to produce bile immediately. Thick, dark, golden-brown / green bile is an indicator of good liver function.

Lactate:-

Patients with pre-operative fulminant hepatic failure/necrosis should have lactate levels that trend towards normal after transplantation providing their fluid status is adequate. Elevated lactate levels also are seen in hypoperfusion states, late sepsis or primary non function.

Post-Operative Considerations:-

A. Hemodynamics:-

(1)Vasodilation: followed by hypotension may occur during the warming phase of hypothermia.

Hypovolemia: may be experienced by the patient due to third spacing of fluids, hemorrhage, sepsis, low oncotic pressure due to hypoalbuminemia, or complement

(1)activation after liver graft reperfusion. This may be treated with volume expanders such as crystalloids NS= 0.9% (NaCl), albumin, or other blood products.

(2)Hypertension: may be due to physiologic stress, shivering, anxiety, pain, hypothermia, tacrolimus or cyclosporine therapy, or fluid overload in patients with renal failure. post-operative hypertension may be treated with a Nipride drip.

Post-Operative Considerations:-

Renal Status:-

Patients who experience renal insufficiency/ failure pre-operative in conjunction with ESLD have what is known as hepatorenal syndrome. Risk factors for continued post-op renal problems

1. Nephrotoxic drugs: Cyclosporine, Tacro, Abelcet, aminoglycosides
2. Intra /post-operative instability coupled with hypoperfusion.
3. Recurring hepatorenal syndrome related to primary nonfunction or acute liver failure.

Post-Operative Considerations:-

C. Laboratory Values:-

Hyponatremia: Remobilization of free water.

Tx: be corrected slowly with fluid restriction.

Hypernatremia: Occurs with dehydration, diuretics, salt-rich solutions (albumin).

Tx: Rehydrate, decreased/discontinue diuretics, mix albumin in D5W instead of the usual NS.

Hypokalemia: Seen with diuretic usage, metabolic alkalosis, insulin drip therapy, steroid therapy, hypothermia,

Tx: potassium supplements.

Hyperkalemia: Occur as result of renal failure, primary non function of Graft (PNFG), multiple blood transfusions, or as a side effect of Tacro or Cyclosporine.

Tx: If acutely high, may require insulin / glucose regimen, Kay oxlate, or dialysis.

Post-Operative Considerations:-

Hypomagnesemia: Related to hypocalcemia, Cyclosporine/Tacro side effects.

Tx: Magnesium supplements.

Hypocalcemia: May be the result of massive blood transfusions.

Tx: Calcium supplements.

Hypophosphotemia: Weakens respiratory muscles, adversely affecting ventilator weaning. Causes neuromuscular irritability and contributes to platelet dysfunction.

Tx: phosphate supplements.

Post-Operative Considerations:-

Hyperphosphotemia: Related to renal failure.

Tx: Phosphate binders (CaCO_3), or dialysis

Hypoglycemia: Recurrent acute liver failure.

Tx: Dextrose IV may be necessary based on blood glucose levels.

Hyperglycemia: Post-operative is considered a good sign of glycogen storage and gluconeogenesis by the grafted liver. May also result from steroid therapy or sepsis later in the post-operative phase.

Tx: Insulin drip per protocol or sliding scale.

Complications of liver transplantation:-

Two of the most common complications following liver transplantation are rejection and infection.

Rejection:-

Immune system works to destroy foreign substances that invade body. The immune system, however, cannot distinguish between transplanted liver and unwanted invaders, such as viruses and bacteria. Therefore, your immune system attempts to attack and destroy new liver.

- **Types of rejection:-**

There are 4 types of rejection that can occur

Hyper acute rejection:-

Occurs minutes to hours after transplantation secondary to presensitization (Action of B-cells) this process Acute and rare. Total hepatic failure results and immediate retransplantation is required.

Accelerated or sub acute rejection:-

Occurs within 2-4 days post operation secondary to (Action of T-cells) it is recognized by fever, jaundice, decreasing quality and quantity of bile, malaise, clay-colored stool.

Types of rejection:-

Acute rejection:-

Occurs in 7 days to several weeks post operative. Secondary to T-cell action, with the same symptoms as accelerated rejection.

Chronic rejection:-

Occurs months to years post-operative due to action of a combination of T-cells and B-level of ALK. phos, GGT, AST, ALT, and bilirubin are seen.

Complications of liver transplantation:-

Infection:-

Because antirejection drugs that suppress immune system are needed to prevent the liver from being rejected, are at increased risk for infections. This problem diminishes as time passes. Not all patients have problems with infections, and most infections can be treated successfully as they occur.

